Competing Bandits in Matching Markets
An Experimental Reproduction and Extension Study

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Abstract

This report reproduces and extends the main results of *Competing Bandits in Matching Markets* (Liu, Simchi-Levi, and Wang, 2020), which studies how bandit learning interacts with stable matching mechanisms. We first re-implement the core components of the model, including classical MAB algorithms (ETC and UCB) and the Gale–Shapley stable matching algorithm. We then reproduce the paper's centralized ETC and UCB experiments, covering key examples that illustrate gap dependence, linear regret phenomena, and heterogeneous outcomes in large markets. Finally, we explore three extensions: a structural optimal-matching regret metric, experiments on two-sided random markets, and a two-sided ETC scheme in which both agents and arms learn preferences. Together, these results provide a comprehensive empirical analysis of learning and competition in matching markets, highlighting both the robustness and the limitations of centralized ETC and UCB in structured and unstructured environments.

1. Introduction

Classical multi-armed bandit (MAB) problems model a single decision-maker who repeatedly chooses among a set of actions (arms) with unknown reward distributions, balancing exploration and exploitation. However, in many real-world markets, there are *many* decision-makers competing for limited resources, and these resources may also have preferences over who uses them. Examples include online labor markets where employers learn which workers they prefer while workers choose between tasks, or college admissions where applicants and universities evaluate each other.

Liu et al. (2020) study this setting through the framework of *competing bandits in matching markets*. In their model, several learning agents (players) compete for arms, and if multiple players choose the same arm, only one is served according to the arm's ranking. The goal is to reach a (player-optimal) *stable matching* between players and arms, while each player still faces a bandit-style exploration–exploitation problem.

The authors analyze centralized and decentralized algorithms. In the centralized setting, a platform collects reported rankings from players and computes matchings using the Gale–Shapley (GS) algorithm. Based on this mechanism, they propose centralized Explore-Then-Commit (ETC) and centralized UCB algorithms with regret guaranties defined relative to stable matchings. They also introduce a decentralized ETC scheme in which players act independently.

In this seminar project, we reproduce several of the original experiments and explore how these algorithms behave beyond the settings of the paper. Our contributions are as follows.

- Providing an intuitive explanation of the model, the centralized ETC and UCB algorithms, and the regret notions based on stable matchings.
- Reproducing three key examples from the paper (Examples 2, 6, and 7) and comparing the regret behavior among players.
- Implementing three extensions: (i) an *optimal matching regret* metric that evaluates structural accuracy, (ii) experiments on random two-sided markets, comparing centralized UCB and ETC, and (iii) a two-sided ETC scheme in which both players and arms learn their preferences.

Our findings support the main insight of Liu et al. (2020): the interaction between bandit learning and stable matching leads to complex dynamics and can produce unexpected failures. At the same time, our extensions show that the good performance seen in simple settings does not always carry

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over to realistic or unstructured markets, where learning can become unpredictable.

Organization of the report. The rest of the report is organized as follows. Section 2 reviews the necessary background on multi-armed bandits and stable matching. Section 3 presents the competing bandits framework and the centralized algorithms from the original paper. Section 4 reproduces the key UCB experiments of Liu et al. (2020). Section 5 introduces and evaluates our three extensions. Section 6 discusses broader implications, and Section 7 concludes.

2. Background and Basic Concepts

2.1. Single-player multi-armed bandits

In the standard stochastic K-armed bandit problem, a single agent repeatedly chooses arms $a \in \{1, \dots, K\}$. When arm a is played in the round t, the agent receives a noisy reward $X_t(a)$ with an unknown mean μ_a. The goal is to design a policy that maximizes cumulative expected reward or, equivalently, minimizes *regret* relative to always pulling the best arm in hindsight.

Two classical algorithms are:

Explore-Then-Commit (ETC). ETC splits the time into an exploration phase and an exploitation phase. In exploration, the agent pulls each arm a fixed number h of times and estimates its mean. In exploitation, it commits to the empirically best arm forever. Intuitively, ETC "learns once" and then stops exploring. This makes it simple, but brittle: early noise can lead to committing to the wrong arm, and it requires knowing the horizon n and a gap parameter to choose h properly.

UCB (Upper Confidence Bound). UCB algorithms maintain for each arm an empirical mean and an exploration bonus that shrinks as the arm is played more often. A typical UCB index has the form

$$\mathrm{UCB}_a(t) = \hat{\mu}_a(t) + \sqrt{\frac{c \log t}{N_a(t)}},$$

where $N_a(t)$ is the number of times that arm a has been played so far. In each round, the agent pulls the arm with the largest index. UCB adaptively balances exploration (trying uncertain arms) and exploitation (preferring arms with high estimated mean), and achieves logarithmic regret in the single-player setting.

Importantly, both ETC and UCB are designed for a *single agent* who does not compete with others. In the matching-market setting, many agents share arms, and conflicts are resolved using the arms' preferences.

2.2. Stable matching and Gale–Shapley

Stable matching is a classical model for two-sided markets, such as college admissions or job markets. There are a set of agents (e.g., students) and a set of arms (e.g., universities). Each agent has a preference ordering over arms, and each arm has a preference ordering over agents.

A matching is a mapping that assigns each agent to at most one arm (and vise versa). A pair (p, a) is called a *blocking pair* for matching m if both p and a strictly prefer each other to their partners in m (or are unmatched). A matching is *stable* if it has no blocking pair.

The Gale–Shapley (GS) algorithm computes a stable matching via a deferred-acceptance process:

1. Each agent proposes to their favorite arm that has not yet rejected them.
2. Each arm keeps its favorite proposal according to its preferences and rejects the rest.
3. Rejected agents propose their next choice, and the process repeats until no more rejections occur.

When agents are the proposing side, GS outputs the *agent-optimal* stable matching: every agent receives its best possible partner among all stable matchings, while arms receive their pessimal stable partners. This asymmetry is central to defining regret notions in the matching bandit model.

In traditional matching theory, it is assumed that both sides know their complete preference lists. In our setting, this is unrealistic: players must gradually learn their preferences through repeated interactions and noisy rewards. Gale–Shapley is then used not on the true preferences but on the current *estimated* preferences induced by bandit algorithms.

3. Competing Bandits in Matching Markets

3.1. Problem setting

Following Liu et al. (2020), we consider N agents (players) $p_1, \dots, p_N$ and K arms $a_1, \dots, a_K$, with $N \le K$. In each round t, every agent p_i selects an arm $m_t(i) \in \{1, \dots, K\}$. If multiple agents choose the same arm a_j, the arm accepts exactly one agent—its most-preferred one according to a fixed ranking π_j over agents—and the others receive reward zero.

If agent p_i successfully matches with arm a_j at round t, they receive a stochastic reward $X_{i,j}(t)$ drawn from a 1-sub-Gaussian distribution with mean $\mu_{i,j}$. In our experiments, for simplicity and following the implementation style of the original paper, we instantiate this model with Gaussian rewards $\mathcal{N}(\mu_{i,j}, 1)$. The vector $(\mu_{i,1}, \dots, \mu_{i,K})$ induces the true preference ordering of player i over arms: $a_j \succ_i$

$a_{j'}$ iff $\mu_{i,j} > \mu_{i,j'}$. Arms have fixed known rankings π_j over players; players *do not* know their own preferences in advance and must learn them through rewards.

Given the true preferences on both sides, GS yields a player-optimal stable matching $\overline{m}$ and an arm-pessimal stable matching $\underline{m}$. The learning problem is to design algorithms for players (and possibly a central platform) so that over time the induced matchings m_t approach stability, while players learn which arms they like.

3.2. Centralized platform and rankings

The paper focuses first on a *centralized* setting with a platform. Each round, every agent submits a ranking $r_{i,t}$ of the arms to the platform. The platform, knowing the arms' fixed rankings $\{\pi_j\}$, runs GS on the reported rankings and outputs a stable matching m_t. Players then observe their own match and reward (if any) and update their estimates.

Two platforms are considered:

- **Explore-Then-Commit platform:** The platform first enforces a structured exploration phase where agents are cyclically rotated over arms; afterwards it performs a single GS call and commits to that matching forever.
- **Gale-Shapley (UCB) platform:** Every round, UCB is run on the current player rankings. Agents are free to change their ranking over time (e.g., based on UCB indices).

In both cases, agents cannot coordinate or see each other's rewards; they only see their own.

3.3. Stable-matching-based regret

The novelty of the model is that regret is defined with respect to stable matchings rather than a single best arm. For player i, let $\overline{m}$ be their agent-optimal stable arm and $\underline{m}(i)$ their agent-pessimal stable arm. The paper defines:

$$R_i^{\text{opt}}(n) = n\mu_{i,\overline{m}(i)} - \mathbb{E}\Big[\sum_{t=1}^{n} X_{i,m_t(i)}(t)\Big], \qquad (1)$$

$$R_i^{\text{pes}}(n) = n\mu_{i,\underline{m}(i)} - \mathbb{E}\Big[\sum_{t=1}^{n} X_{i,m_t(i)}(t)\Big], \qquad (2)$$

the *agent-optimal stable regret* and *agent-pessimal stable regret*. In general, algorithms that are favorable to the proposing side may not achieve sublinear agent-optimal regret, but can still achieve logarithmic agent-pessimal regret.

3.4. Centralized ETC

Centralized ETC uses the Explore-Then-Commit platform. In the exploration phase (lasting hK rounds), the platform ensures that each agent pulls each arm exactly h times by cycling matchings. Agents update their empirical means $\hat{\mu}_{i,j}$ and, after exploration, submit rankings consistent with these means. The platform then runs GS once and commits to the resulting matching forever.

The authors show that with an appropriate choice of h (depending on the unknown gaps and horizon), centralized ETC achieves $O(\log n)$ agent-optimal stable regret but at the cost of requiring knowledge of the minimum gap and horizon. Moreover, the regret of one agent depends on gaps of *other* agents, which is a purely multi-agent phenomenon.

3.5. Centralized UCB

Centralized UCB uses the GS platform. Each round, agent i computes a UCB index $u_{i,j}(t)$ for each arm j, using their own past observations. They rank arms in decreasing order of $u_{i,j}(t)$ and submit this ranking. The platform then runs GS and returns a stable matching w.r.t. these UCB-induced preferences and the fixed arm rankings.

The main result is a problem-dependent $O(\log n)$ upper bound on agent-pessimal regret $R_i^{\text{pes}}(n)$ for every player. However, the paper also constructs examples (Example 6) where centralized UCB *fails* to achieve sublinear agent-optimal regret: some players get stuck in suboptimal but stable configurations because they never get another chance to correct early mistakes.

Finally, the authors show that, under certain conditions, truthful reporting of UCB-based rankings is incentive compatible: a single agent cannot significantly improve their stable regret by deviating from the UCB ranking if all others use UCB.

3.6. Related work

Our work is built directly on Liu et al. (2020), who first introduced the competing bandits in the matching markets framework and analyzed the centralized ETC and UCB algorithms under stable-matching-based regret notions. More broadly, the problem relates to stochastic matching and online allocation under uncertainty, as well as multi-agent bandit models where several learners interact through a shared environment. There is also a growing literature on learning-to-match problems, where recommender systems or platforms must learn preferences on both sides of a market and ensure stable or fair outcomes. Our contribution is purely empirical: we reproduce key experiments of Liu et al. (2020) and explore how their mechanisms behave in more realistic or unstructured markets.

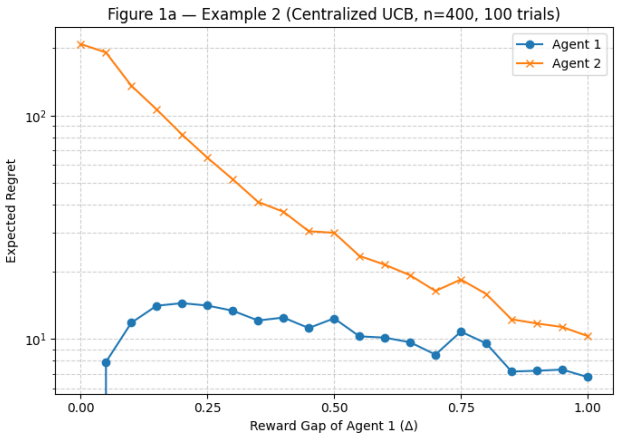

Figure 1. Reproduction of Example 2: expected regret of both agents under centralized UCB as a function of the gap Δ.

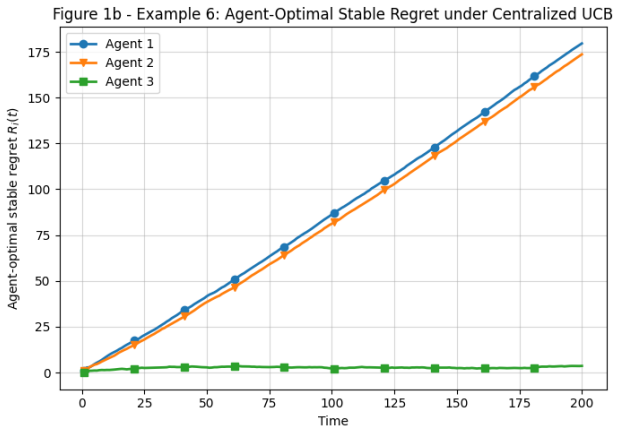

Figure 2. Reproduction of Example 6: agent-optimal regret under centralized UCB. Two agents suffer essentially linear regret.

4. Reproducing the UCB Experiments

We now describe our reproduction of three key examples from the original paper (GitHub Repository, 2025). Our code closely follows the model and algorithms described above.

Unless otherwise stated, rewards are Gaussian with variance 1, horizons and the number of trials match the original experiments, and we report average regret over trials. Our focus is on qualitative behavior across players rather than precise numerical values. We emphasize that our goal is qualitative reproduction of the main phenomena rather than an exact numerical replication of every curve in the original paper.

4.1. Example 2: dependence on the reward gap

Setup. Example 2 studies a minimal market with $N = 2$ agents and $K = 2$ arms. Agent p_1 prefers a_1 over a_2, with mean rewards $(\Delta, 0)$; agent p_2 prefers a_2 over a_1, with mean $(0, 1)$. Both arms prefer p_1 over p_2. The unique stable matching is (p_1, a_1) and (p_2, a_2).

We run a centralized UCB for the horizon $n = 400$ and vary the gap parameter $\Delta \in [0, 1]$. For each Δ, we simulate 100 independent runs and compute the "slot-arm" regret reported in the paper: regret with respect to the arm of the same index (which coincides with the stable matching here).

Results. Figure 1 shows our reproduced curves. For small gaps Δ, agents have difficulty distinguishing their preferred and non-preferred arms; the exploration phase lasts longer, leading to greater regret for both agents. As Δ increases, discrimination becomes easier and regret decreases, roughly matching the trend in Figure 1a of the original paper.

Overall, our curves qualitatively match those in the paper: smaller gaps lead to higher regret, confirming that centralized UCB inherits the gap dependence from single-agent UCB but modified by competition.

4.2. Example 6: linear agent-optimal regret

Setup. Example 6 constructs a 3-agent, 3-arm market where centralized UCB fails to achieve sublinear agent-optimal regret. The preference structure is such that if player p_3 mis-orders a_1 and a_3 once, the GS platform can lock the system into a stable matching where p_3 never again gets to pull a_1 and thus never corrects its mistake.

We replicate the configuration from the paper (true preferences and mean rewards) and run centralized UCB for $n = 200$ rounds over 100 trials.

Results. Figure 2 plots the agent-optimal regret of the three players over time. Agent p_3 has small regret because its best stable partner is clear, but agents p_1 and p_2 suffer regret that grows roughly linearly in n, in line with Figure 1b in the original paper. This confirms that centralized UCB can converge to a stable matching which is not agent-optimal for some players, and that they may never get a chance to escape even with ongoing exploration.

4.3. Example 7: global preferences and heterogeneous regret

Setup. Example 7 considers a "global preference" structure with $N = K = 20$, where all agents have the same ranking over arms and all arms have the same ranking over agents. The unique stable matching is (p_i, a_i) for $i = 1, \ldots, 20$, and mean rewards decrease uniformly across arms.

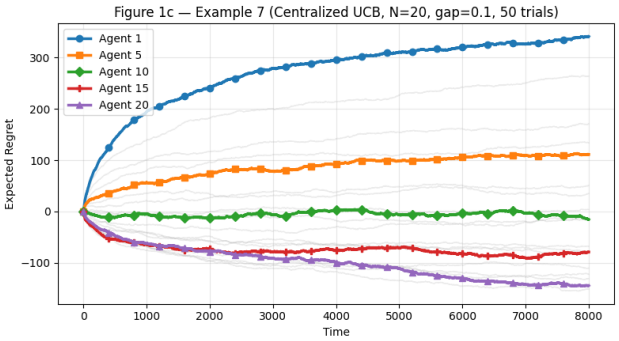

Figure 3. Reproduction of Example 7: agent-optimal regret for selected agents under centralized UCB in a global preference market.

Table 1. Summary of the experimental setups used in the extensions.

Extension	(N, K)	T	Trials
A: Optimal-matching regret	(2,2)	400	100
B1: Random instance generator	(6,6)	500	20
B2: UCB vs. ETC	(6,6)	500	20
C: Two-sided ETC	(4,4)	800	–

We run centralized UCB for $n = 8000$ rounds and average over 50 trials.

Results. Figure 3 plots the agent-optimal regret for selected agents. Top-ranked agents (e.g., p_1) pay a large price when they deviate from their ideal arm because all other arms are significantly worse; they exhibit positive but sublinear regret. Mid-ranked agents have near-zero regret on average: sometimes they get better-than-stable arms, sometimes worse, roughly canceling out. Low-ranked agents often benefit from exploration, as they occasionally obtain arms above their pessimistic stable partner, and thus can exhibit negative regret when measured against the pessimal benchmark.

Our curves qualitatively match Figure 1c in the paper and illustrate a key phenomenon: in these markets, learning by some agents can systematically benefit or hurt others.

5. Extensions

Beyond reproducing the original experiments, we also examine how the algorithms behave in settings not covered by Liu et al. (2020). We introduce the three extensions and evaluate their behavior. These extensions test how well the mechanisms perform when the assumptions are relaxed or when both sides of the market must learn.

5.1. Extension A: optimal-matching regret

The original regret notions mix reward quality and market structure: they compare the observed rewards to the utilities obtained under agent-optimal or agent-pessimal stable matchings. To purely measure *matching accuracy*, we introduce *optimal-matching regret*:

$$R_{\text{opt}}^{\text{match}}(T) = \sum_{t=1}^{T} \mathbf{1}\{m_t \neq m^\star\}, \tag{3}$$

where m_t is the matching produced in round t and $m^\star$ is the agent-optimal stable matching under true preferences. This regret ignores actual reward values and only counts how often the algorithm outputs an incorrect matching.

We compute $R_{\text{opt}}^{\text{match}}(t)$ for centralized UCB and centralized ETC in Example 2 with a fixed gap $\Delta = 0.5$. We run each algorithm for a horizon of $n = 400$ rounds and average over 100 independent trials, using the same Gaussian reward model as in reproduction experiments.

Results. Figure 4 shows that the centralized UCB's matching regret grows initially as it explores, but then stabilizes: the algorithm tends to converge to $m^\star$ and stays there, only occasionally deviating. In contrast, centralized ETC spends a long initial phase cycling through all matches before committing. During this exploration period, it repeatedly outputs non-optimal matchings, which causes $R_{\text{opt}}^{\text{match}}(t)$ to increase almost linearly. Once the algorithm commits, it freezes on the learned matching, and the regret curve becomes flat—meaning it cannot correct any mistakes made during exploration.

This extension highlights a structural difference: UCB is adaptive and self-correcting, while ETC is brittle. Even when their standard reward-based regrets might look similar in some settings, the matching-regret perspective reveals persistent structural errors for ETC.

5.2. Extension B1: Random instance generator

To investigate the behavior of centralized UCB in unstructured matching markets, we implement a generator for arbitrary two-sided instances. Given a number of agents N, a number of arms K, and a reward-gap interval $[\Delta_{\min}, \Delta_{\max}]$, the generator constructs a random yet internally consistent market through the following steps:

- **Random preference orders:** each agent samples a uniform random permutation of the K arms, and each arm samples a uniform random permutation of the N agents.
- **Consistent mean rewards:** for each agent i, the arm at the bottom of their ranking receives the mean reward

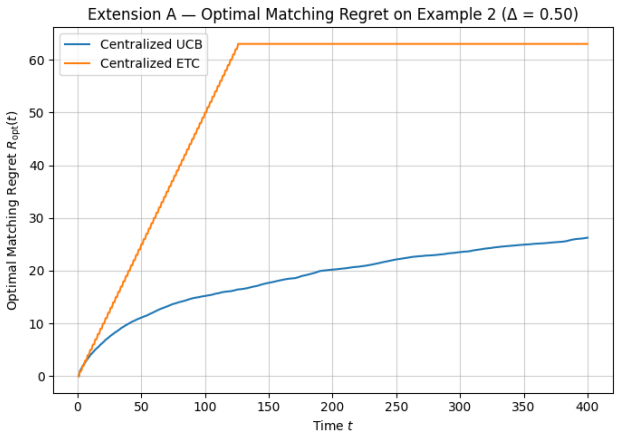

Figure 4. Extension A: optimal-matching regret $R_{\text{opt}}^{\text{match}}(t)$ for centralized UCB vs. centralized ETC on Example 2.

0, and moving upward adds an independent random gap in $[\Delta_{\min}, \Delta_{\max}]$. Arm-side utilities $\gamma_{j,i}$ are constructed analogously. This guaranties that the generated mean rewards strictly reflect the sampled preferences.

Using this generator, we run centralized UCB in 20 independently sampled markets with $N = K = 6$, horizon $T = 500$, and reward gaps in $[0.2, 0.5]$. For each trial, we record the best-arm regret of each agent, then average across trials. The resulting regret trajectories are shown in Figure 5.

Results. These random-market experiments behave very differently from the clean, structured examples studied earlier:

- For most agents, regret grows almost linearly, showing that UCB struggles to consistently find or keep the best arm.
- Agents experience very different outcomes: some are matched with good arms often, while others repeatedly lose conflicts and accumulate much higher regret.
- These differences come entirely from random variations in preferences and priorities, and nothing in the setup favors any particular agent.

These results show that centralized UCB strongly depends on the structure of the market. It performs well in simple, well-organized settings, but in random and noisy environments, it cannot reliably resolve conflicts or reach stable, low-regret outcomes.

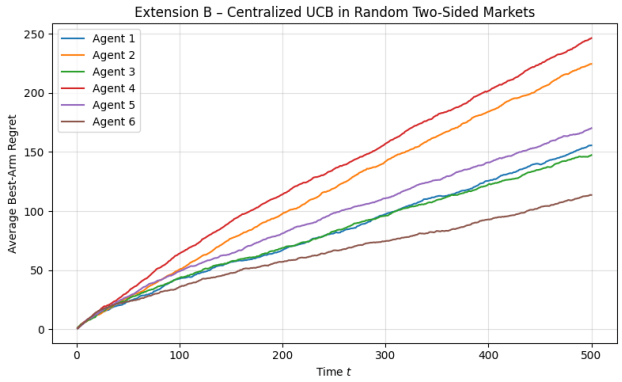

Figure 5. Extension B1: Average best-arm regret under centralized UCB in randomly generated two-sided markets ($N = K = 6$, $T = 500$, 20 trials). Each curve corresponds to one agent, averaged over independently sampled preference profiles and consistent reward structures. Regret grows roughly linearly for most agents, with substantial variation driven purely by random asymmetries in preferences and arm priorities.

5.3. Extension B: random two-sided markets and UCB vs. ETC

The experiments in the original paper use carefully constructed preference structures. To study more realistic scenarios, we reuse the Extension B1 random instance generator, which produces two-sided markets with independently sampled preference lists and mean rewards consistent with those preferences on both sides.

Concretely, we consider markets with $N = K = 6$, horizons $T = 500$ rounds, and 20 independent trials per configuration. For each pair of agent-arm (i, j), we sample a mean $\mu_{i,j}$ by first drawing a random total order over the arms for each agent, then assigning the base value 0 to the least-preferred arm and adding gaps drawn uniformly from $[\Delta_{\min}, \Delta_{\max}] = [0.2, 0.5]$ along the ranking.

Results. In these unstructured markets, both algorithms struggle:

- Centralized ETC is often catastrophic: if the exploration horizon h is not tuned exactly, many agents commit to clearly suboptimal arms. Matching-regret and reward-regret both grow nearly linearly after the commit phase.
- Centralized UCB performs better, but still exhibits steadily growing regret for many agents. Conflicts are frequent and slight asymmetries in preferences cause some agents to "win" most conflicts and others to be systematically disadvantaged.

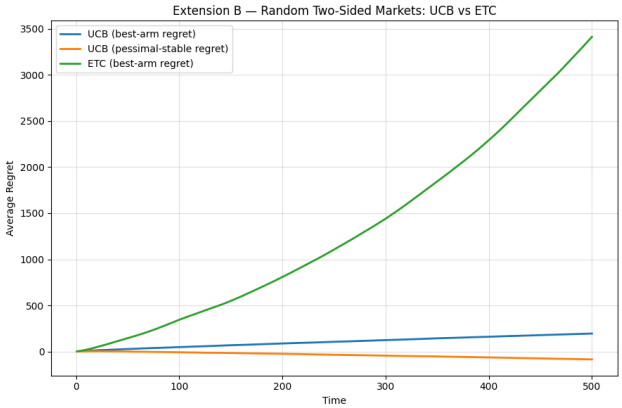

Figure 6. Extension B: regret of centralized UCB and ETC in random two-sided markets.

Representative curves are shown in Figure 6. The favorable guaranties and behaviors observed in structured examples do not easily transfer to random markets. The interaction of bandit learning with competition and random preferences leads to unstable learning dynamics and inequality across agents: some consistently end up in good matches, others in poor ones.

5.4. Extension C: Two-Sided ETC

Real-world matching platforms often exhibit uncertainty on *both* sides: workers learn which firms suit them, and firms simultaneously learn which workers perform well. To explore the consequences of such bilateral uncertainty, we study a two-sided variant of ETC in which both players and arms act as bandit learners. In the original model, arms have fixed known rankings over players; here, when an arm is matched with an agent, it observes a noisy reward and updates an estimate of its preferences. This mirrors settings such as firms that evaluate applicants, universities that evaluate students, or platforms that match buyers and sellers.

We implement a two-sided centralized ETC algorithm with the following structure:

- **Exploration phase:** the platform cycles through all player–arm pairs so that each pair (i, j) is matched exactly h times. Both players and arms observe Gaussian rewards and update empirical means $\hat{\mu}_{i,j}$ and $\hat{\gamma}_{j,i}$.
- **Preference formation:** after exploration, each player ranks the arms decreasing $\hat{\mu}_{i,j}$, and each arm ranks the players decreasing $\hat{\gamma}_{j,i}$.
- **Commit phase:** Gale–Shapley is run once on these learned rankings, and the resulting matching is played forever.

Choosing the exploration length h. In one-sided ETC, the recommended exploration length is

$$h = \left\lceil \frac{4}{\Delta^2} \log\left(1 + \frac{n\,\Delta^2 N}{4}\right) \right\rceil,$$

where Δ is the smallest agent-side gap.

For the two-sided case, both agents and arms must correctly identify their stable partners. We therefore define the agent-side and arm-side gaps:

$$\Delta_{\text{agents}} = \min_{i,\,j:\,\mu_{i,m^\star(i)} > \mu_{i,j}} \big(\mu_{i,m^\star(i)} - \mu_{i,j}\big),$$

$$\Delta_{\text{arms}} = \min_{j,\,i:\,\gamma_{j,n^\star(j)} > \gamma_{j,i}} \big(\gamma_{j,n^\star(j)} - \gamma_{j,i}\big).$$

Plugging these values for Δ into the equation for h, we then compute the corresponding exploration requirements h_{agents} and h_{arms}.

The two-sided ETC exploration length is then

$$h \;=\; \max\{\, h_{\text{agents}},\; h_{\text{arms}} \,\}.$$

In our 4×4 market with horizon $n = 800$ and the gaps used in our notebook, this yields $h \approx 3044$, which is larger than the entire time horizon.

This inflation of h arises because both sides must correctly rank all alternatives for the true stable matching to be recovered: even a single misordering can change the stable partner of an agent or an arm. As a consequence, the effective gap-dependent sample complexity is amplified, and the exploration budget required by this two-sided ETC scheme quickly exceeds any reasonable time horizon.

Results. Because $h > n$ in this setting, the algorithm never leaves the exploration phase. Players and arms cyclically rotate over all matches without ever exploiting their learned preferences. As a result, regret grows almost linearly for all agents, as shown in Figure 7. There is no stabilization, and the learned rankings remain dominated by noise.

This demonstrates that naive two-sided generalizations of ETC are impractical: the exploration requirements scale extremely poorly with the gaps, and the algorithm fails to reach meaningful exploitation. Designing viable two-sided learning mechanisms likely requires continuous-learning algorithms (e.g., UCB-style updates) or adaptive stopping rules that avoid pathological exploration budgets.

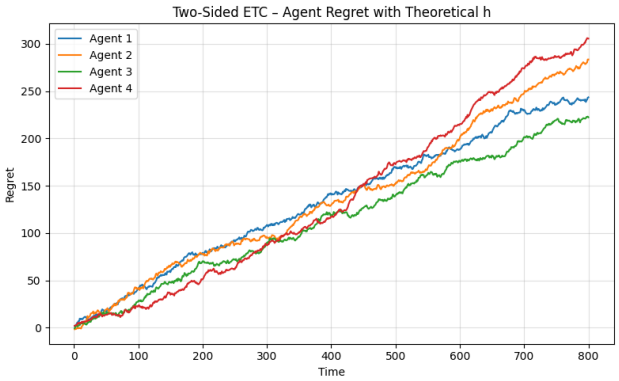

Figure 7. Extension C: regret of a two-sided ETC scheme in a 4×4 market. The commit phase is never reached.

6. Discussion

Market structure matters. In structured environments—such as global preference models or small, hierarchical markets—centralized UCB performs relatively well, often yielding sublinear regret and convergence to the intended stable matching. In contrast, Extensions B1 and B2 show that in randomly generated two-sided markets, conflicts are frequent, preference cycles arise, and small asymmetries can produce highly unequal learning outcomes. Stability alone does not guarantee good regret or fairness in unstructured settings.

ETC vs. UCB. ETC remains brittle across all experiments: a mis-specified exploration length or early noise can lead to irreversible commitment to suboptimal matchings. Extension A illustrates this clearly under optimal-matching regret, where ETC accumulates persistent structural errors. UCB adapts more effectively, but Example 6 shows that it can still become trapped in suboptimal stable matchings when the GS platform restricts future exploration.

Unstructured markets amplify instability. Extension B shows that both centralized ETC and UCB perform poorly in heterogeneous markets. Regret often grows nearly linearly, and agents experience persistent inequalities: some systematically win conflicts, while others incur repeated losses. These outcomes highlight that strong performance in stylized examples does not reliably extend to realistic, noisy markets.

Two-sided learning is difficult. Introducing uncertainty on both sides significantly increases complexity. In Extension C, the exploration requirement for two-sided ETC exceeds feasible horizons, preventing the algorithm from ever entering the commit phase. Regret remains linear, and no stable behavior emerges. Effective two-sided mechanisms likely require continuous-learning approaches rather than commit-style schemes.

Trustworthiness considerations. These findings raise concerns regarding fairness and robustness. Agents may benefit or suffer solely due to randomness or structural asymmetries, and outcomes can vary widely across market configurations. Practical deployment of learning-based matching mechanisms must therefore account for fairness, stability, and interpretability, not only aggregate regret.

7. Conclusion

We revisited the matching bandit model of Liu et al. (2020), connecting it to classical MABs and stable matching, reproducing the key UCB experiments, and exploring three extensions: optimal-matching regret, random two-sided markets, and a two-sided ETC variant.

Our results support the main insights of the original work while emphasizing several limitations. Centralized UCB performs well in structured settings but can break down in unstructured markets; ETC remains brittle and sensitive to early noise; and naive two-sided ETC becomes impractical due to excessive exploration requirements.

Overall, developing matching algorithms that learn continuously, remain fair, and perform robustly across diverse market structures remains an open challenge. Promising future directions include fully two-sided UCB-style mechanisms, fairness-aware objectives, and approaches that ensure stability and robustness in more realistic environments.

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